Diagnostic Results (Data Release)

Specification Echo in Duplicate-Code Measurement of LLM-Generated Classes: A Measurement-Validity Audit on ClassEval

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This document collects the diagnostic results that the manuscript summarizes. Unless stated otherwise, every estimate refers to the task class: detection, Stmt (the number of AST statement nodes, docstrings excluded) and all smells restricted to the class the ClassEval task requests (manuscript, Section 3.2), 1,081 AI and 100 reference classes. "Whole-file configuration" marks the earlier setting, as in manuscript Table 7: the detector scanned the whole generated file, and Stmt and the other six smells were taken from the file's largest class (1,083 AI classes). It differs from the "detection only" setting of manuscript Table 7, which changes the detection scope alone. DP: the documentation-preserving rule; SA: the statement-aware rule; policy contrast: the size-adjusted AI/reference ratio under DP divided by that under SA.

S1 Dense core of the common size range

Table S1: Size-adjusted models refitted on the dense core of the common size range (the reference classes' 10th to 90th percentile, 18–47 Stmt; 568 AI and 81 reference task classes). Ratio: standardized AI/reference ratio of expected total smell counts. Policy: DP ratio over SA ratio with 95% percentile interval from task-cluster bootstrap refits; Rep.: replicates in which both fits converged, of 300. Last column: the same policy contrast in the whole-file configuration (573 AI and 81 reference classes). Source: `size_sensitivity.py`, item 4.3.

Lik.	Model	DP ratio	SA ratio	Policy	Rep.	Whole-file config.
Poisson	M0	2.13	1.82	1.17 [1.05, 1.34]	299	1.24 [1.10, 1.40]
Poisson	Mlin	2.16	1.82	1.19 [1.07, 1.37]	300	1.26 [1.12, 1.44]
Poisson	M1	2.18	1.79	1.21 [1.05, 1.38]	299	1.29 [1.14, 1.48]
NB2	M0	2.13	1.81	1.18 [1.05, 1.36]	300	1.27 [1.10, 1.47]
NB2	Mlin	2.19	1.80	1.22 [1.07, 1.39]	300	1.29 [1.13, 1.45]
NB2	M1	2.13	1.81	1.18 [1.04, 1.35]	300	1.27 [1.09, 1.48]

Inside the dense core the model-based AI/reference ratios are near two under both rules. These are point estimates from the models, reported without intervals here; they suggest, but do not establish, that AI task classes of similar size carry more smells than the reference. The rule change, which has intervals, is between $1.17\times$ and $1.22\times$.

S2 Bootstrap refits

Failed and successful resamples have similar composition (for full-range Poisson M0, about 63 distinct tasks and 17 to 18 reference classes with a smell in both). Including the non-converged fits moves no interval bound by more than 0.04. This shows that the reported intervals are not sensitive to the inclusion criterion; it does not show that the non-converged fits carry no information.

Under Poisson M1 over the common range, the AI/reference rate ratio at the pooled 25th, 50th and 75th percentile of size (17, 23 and 30 Stmt) is $6.60\times$, $3.80\times$ and $2.34\times$ under DP and $4.01\times$, $2.63\times$ and $1.81\times$

Table S2: Outcome of the task-cluster bootstrap refits behind manuscript Table 5. OK: both fits converged (the replicates the policy interval uses); non-conv.: the iteratively reweighted least-squares fit reported non-convergence; error: the fit raised an exception. Interval with non-converged fits included: the policy interval recomputed keeping them. Source: `size_models.py`.

Range	Lik.	Model	OK	Non-conv.	Error	Policy, OK only	With non-conv.
Full	Poisson	M0	774	225	1	1.22 [1.13, 1.37]	[1.12, 1.37]
Full	Poisson	Mlin	1000	0	0	1.29 [1.04, 1.73]	[1.04, 1.73]
Full	Poisson	M1	794	205	1	1.62 [1.17, 2.08]	[1.13, 2.06]
Full	NB2	M0	985	15	0	1.44 [1.18, 1.80]	[1.18, 1.80]
Full	NB2	Mlin	1000	0	0	1.27 [0.97, 1.67]	[0.97, 1.67]
Full	NB2	M1	981	19	0	0.95 [0.31, 1.99]	[0.33, 2.00]
Common	Poisson	M0	500	0	0	1.21 [1.10, 1.36]	[1.10, 1.36]
Common	Poisson	Mlin	500	0	0	1.18 [1.02, 1.36]	[1.02, 1.36]
Common	Poisson	M1	500	0	0	1.53 [1.27, 1.86]	[1.27, 1.86]
Common	NB2	M0	498	2	0	1.40 [1.16, 1.73]	[1.15, 1.73]
Common	NB2	Mlin	498	2	0	1.14 [0.98, 1.40]	[0.98, 1.39]
Common	NB2	M1	499	1	0	1.47 [1.20, 1.94]	[1.20, 1.94]

under SA; these estimates rest on the 17 small reference classes.

S3 Producer heterogeneity

After statement-free blocks are excluded, the tie-corrected Skillings–Mack test over the twelve producers, with tasks as blocks and the SA total smell density as the response, gives 28.44 ($\mathrm{df} = 11$, $p = 0.003$; 100 blocks, 23 complete). Task-matched sign-flip permutation tests ($10{,}000$ permutations) with Benjamini–Hochberg adjustment separate 3 of 66 producer pairs, all involving Gemini-Pro (against DeepSeek-Coder, GPT-4-Turbo and StarCoder-15B; adjusted $p = 0.013$–0.015). In the whole-file configuration the same test gives 33.09 ($p < 0.001$) and 13 of 66 pairs. Source: `producers.py`.

S4 From whole-file to task-class detection

Table S3: What happens to each of the 421 blocks the whole-file detector flags when detection is restricted to the task class. Kept: the identical copy (same physical lines) is flagged in both scopes. Outside: the copy lies wholly outside the task class. First outside: the copy lies inside the task class but its first occurrence does not, so within the class there is nothing for it to repeat. Boundary: the copy straddles the class boundary. No block is flagged only in the task scope. Source: `task_scope_reconcile.py`.

	AI		Reference	
Fate	Blocks	Statement-free	Blocks	Statement-free
Kept	264	198	3	0
Outside	135	96	0	0
First outside	18	8	0	0
Boundary	1	0	0	0
Total	418	302	3	0

The 20 AI files with two or more top-level classes come from SantaCoder-1.1B (13), CodeGeeX2-6B (4) and PolyCoder-2.7B (3); in 12 of them the task class is not the file's largest class, so the whole-file configuration also computed Stmt and the other six smells on a different class. In three files the requested name is defined more than once and the last definition is used. Two CodeGeeX2-6B outputs define no class of the requested name (`DiscountStrategy`, `Warehouse`) and are excluded.

S5 Whole-file configuration estimates

Table S4: Whole-file configuration: size-adjusted AI/reference ratios of expected total smell counts (the counterpart of manuscript Table 5). Columns as there.

		DP		SA			
Lik.	Model	$\hat{\beta}_A$ (SE)	Ratio	$\hat{\beta}_A$ (SE)	Ratio	Policy	OK
Full size range: 1,083 AI, 100 reference classes							
Poisson	M0	0.39 (0.26)	1.47	0.17 (0.24)	1.19	1.24 [1.14, 1.44]	911
Poisson	Mlin	0.47 (0.32)	1.60	0.23 (0.27)	1.26	1.27 [1.07, 1.64]	1000
Poisson	M1	1.86 (0.45)	1.82	1.09 (0.46)	1.25	1.46 [0.69, 2.19]	915
NB2	M0	0.83 (0.22)	2.29	0.47 (0.21)	1.60	1.43 [1.23, 1.75]	977
NB2	Mlin	0.54 (0.31)	1.72	0.40 (0.22)	1.49	1.15 [0.85, 1.58]	971
NB2	M1	2.02 (0.43)	0.47	1.17 (0.44)	1.34	0.35 [0.07, 2.08]	955
Common size range: 828 AI, 100 reference classes							
Poisson	M0	0.38 (0.25)	1.47	0.15 (0.25)	1.16	1.27 [1.14, 1.42]	500
Poisson	Mlin	0.34 (0.27)	1.41	0.12 (0.25)	1.13	1.24 [1.10, 1.43]	500
Poisson	M1	1.53 (0.38)	2.38	1.11 (0.41)	1.43	1.66 [1.40, 1.99]	500
NB2	M0	0.81 (0.23)	2.25	0.46 (0.22)	1.59	1.42 [1.21, 1.67]	498
NB2	Mlin	0.47 (0.25)	1.60	0.36 (0.21)	1.44	1.11 [0.97, 1.42]	475
NB2	M1	1.51 (0.35)	2.29	1.04 (0.38)	1.40	1.63 [1.30, 2.29]	500

Table S5: Whole-file configuration: key estimates on subsets (the counterpart of manuscript Table 6). Columns as there.

Subset	AI n	Tasks	DP	SA	Δdup	Rule change
All classes	1,083	100	3.90	1.00	3.5 [1.9, 5.3]	3.91 [2.82, 5.69]
Without two dominant	883	100	1.09	0.80	1.9 [0.7, 3.2]	1.37 [1.09, 1.67]
Parseable without salvage	831	100	0.90	0.70	1.2 [0.1, 2.2]	1.30 [1.08, 1.64]
All methods non-empty	705	100	1.37	0.97	2.9 [0.9, 5.5]	1.42 [1.11, 1.71]
Passing all tests	190	51	1.71	1.66	1.1 [0.0, 2.5]	1.03 [1.00, 1.12]
Ten others, non-empty	703	100	1.03	0.82	1.8 [0.3, 3.6]	1.25 [1.04, 1.52]
Ten others, incomplete	180	82	1.36	0.71	1.7 [0.4, 3.5]	1.90 [1.00, 2.63]

S6 Size support, statement criterion and size exponents

Table S6: Support inside the common size range: task classes per Stmt band, and how many carry at least one smell under the statement-aware rule. Source: `size_sensitivity.py`, item 4.3.

Stmt band	13–19	20–29	30–39	40–59	60–80
Reference classes	17	40	23	18	2
with a smell	0	5	4	7	2
AI classes	325	295	125	70	7
of which two dominant	74	18	3	2	2
with a smell	12	42	37	30	6

S7 Later cohort and repository check: protocol

For the four checkpoints marked n.r. the snapshot and the numerical precision (full or 4-bit) were not recorded. The generation harness is kept with `transformers` 4.57.6, PyTorch 2.6.0 and `bitsandbytes` 0.49.2; the versions in use on the generation dates were not logged. Because neither the revisions nor the

Table S7: Sensitivity to the statement criterion (task classes, six-line blocks). Kept: blocks retained. AI dup.: AI Duplicate Code blocks/kStmt. Ratio: AI/reference total smell density. Rule change: paired factor relative to the documentation-preserving rule (95% task-cluster interval, 20,000 replicates). A 10% share equals the ≥ 1 statement rule for six lines. Source: `statement_criterion.py`.

Counting rule	Kept	AI dup.	Ratio	Rule change
Documentation-preserving (no requirement)	267	15.36	2.29×	—
≥ 1 statement, def or decorator line	84	2.42	0.82×	2.81× [2.17, 3.79]
≥ 1 implementation statement	69	1.49	0.71×	3.23× [2.43, 4.58]
$\geq 25\%$ implementation-statement lines	67	1.42	0.70×	3.27× [2.45, 4.66]
$\geq 50\%$ implementation-statement lines	66	1.39	0.70×	3.28× [2.46, 4.68]

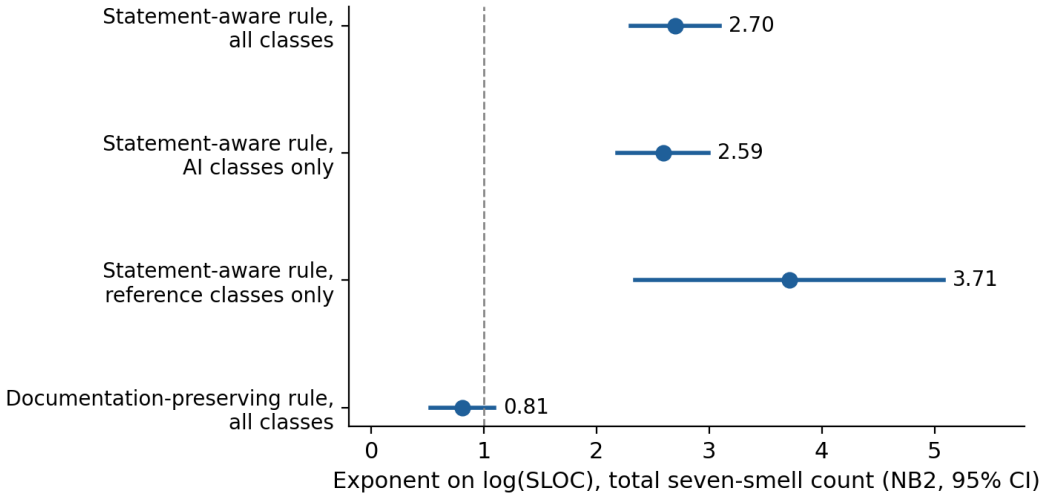


Figure S1: Size exponent of the *total* seven-smell count: the coefficient on $\log(\text{Stmt})$ in an NB2 model with $\log(\text{Stmt})$ as a free covariate (model Mlin), with 95% confidence intervals, under each duplication rule. The dashed line at one is what smells/kStmt assumes. Source: `figure_exponent.py`.

Table S8: Later model-and-prompt cohort (manuscript Section 4.3.8). All six were run locally with Hugging Face `transformers`: ClassEval's holistic instruction ("Please complete the class *name* in the following code." followed by the skeleton) wrapped in each model's own chat template (`apply_chat_template`), greedy decoding (`do_sample=False`), at most 2,048 new tokens, stopping at the model's end-of-sequence token. Outputs were written on 4–5 August 2026. Code was extracted and salvaged exactly as for the published cohort (manuscript Section 3.1). Rev.: Hugging Face snapshot, recorded only where the local cache kept it. Parsed: outputs yielding a parseable class (after salvage); Salv.: of which salvaged; Complete: every requested method has a non-empty body; Pass: all ClassEval tests pass; SF: statement-free blocks. Source: `later_cohort_status.json`.

Checkpoint	Rev.	Parsed	Salv.	Complete	Pass	SF
`Qwen/Qwen2.5-Coder-1.5B-Instruct`	`2e1fd39`	100	3	77	12	4
`Qwen/Qwen2.5-Coder-3B-Instruct`	`488639f`	100	0	100	25	3
`Qwen/Qwen2.5-Coder-7B-Instruct`	n.r.	100	0	99	33	1
`Qwen/Qwen2.5-Coder-14B-Instruct`	n.r.	100	0	100	40	3
`ibm-granite/granite-8b-code-instruct`	n.r.	100	3	91	19	4
`bigcode/starcoder2-15b-instruct-v0.1`	n.r.	100	1	99	28	2
Total		600	7	566	157	17

precision are pinned for every model, this cohort is reported as a check on the published one, not as a reproducible comparison of newer models.

Repository check (manuscript Sections 3.5 and 4.3.9). *Discovery:* the GitHub commit-search API was queried for each of six phrases—`"Co-authored-by: Copilot"`, `"Co-authored-by: Claude"`, `"Co-authored-by: Cursor"`, `"Co-authored-by: Devin"`, `"Generated with Claude Code"` and `"copilot-swe-agent"`—each with `language:Python`, pages 1–3 of 100 results in the API's default (best-match) order; forks were skipped, leaving 654 candidate repositories in discovery order. *Admission:* candidates were processed in that order until 100 repositories had at least eight eligible files in each group. *Attribution:* a file is AI-involved if the commit that added it has a message matching an agent trailer (Copilot, Claude, Cursor, Devin or Codex), and human-attributed if it was added by any other commit after the repository's first such commit; a missing trailer does not establish human authorship. *Files:* non-merge commits on all refs in `git log` order (newest first), the first addition of each `.py` path, paths containing `/test` excluded, and the first 80 files per group whose largest class parses. *Pinning:* the search date was not recorded; at the re-audit every history walk was restricted to commits before 1 August 2026, 97 repositories were still public, and their HEAD commits are released in `repository_heads.json`.

S8 Policy-contrast stability and task-indicator fits

Table S9: Size-adjusted policy contrast (DP ratio over SA ratio) with 95% percentile interval from task-cluster bootstrap refits; Rep.: refits in which both fits converged. Source: `ijseke_stability.py`.

		(a) Common, 2,000 refits		(b) Full, 21 dropped		(c) Common, 16 dropped	
Lik.	Model	Policy	Rep.	Policy	Rep.	Policy	Rep.
Poisson	M0	1.21 [1.09, 1.36]	2000	1.22 [1.13, 1.39]	385	1.22 [1.10, 1.36]	500
Poisson	Mlin	1.18 [1.01, 1.37]	2000	1.26 [1.11, 1.48]	500	1.23 [1.09, 1.42]	500
Poisson	M1	1.53 [1.29, 1.87]	2000	1.56 [1.15, 2.00]	387	1.40 [1.16, 1.75]	500
NB2	M0	1.40 [1.16, 1.76]	1998	1.34 [1.11, 1.77]	499	1.27 [1.09, 1.51]	500
NB2	Mlin	1.14 [0.96, 1.36]	1983	1.31 [1.15, 1.54]	500	1.25 [1.10, 1.42]	500
NB2	M1	1.47 [1.19, 1.93]	2000	1.50 [1.17, 1.89]	493	1.35 [1.13, 1.63]	500

(a) Bootstrap endpoints The common-range models of manuscript Table 5 (500 refits) are refitted with 2,000 refits from a new random stream (seed changed). No interval bound moves by more than 0.04, and the same five of six intervals exclude one.

(b), (c) Influential classes Drop criterion: Cook's distance from the Poisson Mlin fit on all classes of the range, computed separately under each rule; under each rule the classes at or above the 99th percentile are flagged, and the union of the two flagged sets is dropped from the sample used under *both* rules. Full range: 14 classes flagged under DP and 12 under SA, union 21 (17 AI); common range: 10 and 10, union 16 (13 AI). All six models are then refitted under both rules on the reduced sample (point estimates and 500 refits, as in Table 5). For comparison, dropping each rule's own top 1% moves the Poisson Mlin AI/reference ratio from 1.43 to 2.05 (DP) and from 1.11 to 1.80 (SA) over the full range, and from 1.25 to 2.00 and from 1.06 to 1.58 over the common range; these come from different samples and give no policy contrast. Over the full range, Poisson M0 and M1 converge in only 385 and 387 of 500 refits. The analysis checks one form of influence; it does not establish stability under other sample selections or model specifications.

Table S10: Poisson M0 with one indicator per task (manuscript Section 4.3.2): 300 task-cluster refits per range, two rules each. Values are minimum, median, maximum unless stated. Source: `ijseke_taskfe_diag.py`, `ijseke_separation.py`.

	Full range	Common range
All-zero tasks dropped before fitting	21, 44, 67	27, 48, 69
Tasks kept	33, 56, 79	31, 52, 73
Fits (refit × rule) where IRLS stops at 300 iterations	85 of 600	0 of 600
Fits with separation (LP check)	0 of 600	0 of 600
Newton–Raphson converged	600 of 600	600 of 600
The 85 fits IRLS does not finish (Newton–Raphson solution)		
Deviance change over the last 10 IRLS iterations	$1.5 \cdot 10^{-7}$, $3.4 \cdot 10^{-4}$, 0.024	—
Largest absolute coefficient (always a size-spline term)	68, 98, 471	—
Smallest fitted mean	10^{-204}, 10^{-38}, 10^{-29}	—
Smallest eigenvalue of the Fisher information	$1.3 \cdot 10^{-6}$, 0.001, 0.004	—
$\lvert \log \rvert$ difference of the AI ratio, IRLS vs. Newton–Raphson	at most 0.062	—
Policy contrast, median and 95% interval		
IRLS, refits converged under both rules	1.36 [1.08, 1.74] ($n = 224$)	1.20 [0.93, 1.57] ($n = 300$)
Newton–Raphson, all refits	1.32 [1.03, 1.73] ($n = 300$)	1.20 [0.93, 1.57] ($n = 300$)
Newton–Raphson, refits IRLS did not finish	1.21 [0.93, 1.62] ($n = 76$)	—

Zero tasks A task whose classes carry no smell under a rule has an indicator whose maximum-likelihood estimate is $-\infty$ (complete separation); such a task also carries no information on the AI coefficient once task is controlled for. These tasks are dropped before fitting, separately for each rule and refit.

Separation check For a log-link Poisson model the maximum-likelihood estimate fails to exist if and only if some direction $b \neq 0$ satisfies $x_i'b = 0$ for every class with a smell and $x_i'b \leq 0$ for every class without one, strictly for at least one. For every fit we maximized $-\sum_{y_i=0} x_i'b$ under these constraints and $-1 \leq b_j \leq 1$ (HiGHS linear program); the optimum was zero in all 1,200 fits, so the estimate exists in each.

Reading The estimates exist, but in the 85 full-range fits IRLS does not finish, the size spline takes extreme values: the largest coefficient is a spline term every time, and the smallest fitted means are numerically zero. These are the smallest classes: all 3,381 fitted means below 10^{-6} in these fits belong to classes of 2 to 10 Stmt, below the smallest reference class (13), and none carries a smell, so the spline can fit their zero counts almost exactly (`ijseke_separation_tail.py`). The fits are thus close to separation in the size terms; this explains IRLS's slow convergence, while the Fisher information stays positive definite and Newton–Raphson converges in every fit. The AI coefficient is little affected (IRLS and Newton–Raphson differ by at most 6% where IRLS stopped). Over the common range no fit shows this behavior. The task-adjusted estimate is a secondary check, and over the common range its interval includes one.

S9 Reproduction

What is public and what is not is the same as in the manuscript's Data Availability Statement. The derived data are public at `https://github.com/Dongkwan/ai-code-td`: for both scopes, the per-class metrics under both rules, every flagged block with its line labels, the task-class identification of each file, and the stored outputs behind every table here (point estimates, fitted-model summaries and interval bounds). With these data, the counts, densities and ratios in the tables can be checked and re-aggregated without the code, and the reported model estimates and intervals can be compared with the stored outputs. Refitting the count models and re-running the bootstrap from scratch is a different matter: it requires the code, or a reimplementation of the procedure described in the manuscript, and reproducing the intervals exactly also requires the code's random seeds, since the bootstrap replicates themselves are not part of the release. The source code is not public. The scripts named in the captions belong to it; it is available from the corresponding author on request and can be provided to reviewers through the editor. That package runs under Python 3.10.11 with pinned requirements and `jscpd` 4.0.5; its `README.md` gives the order of the scripts, and `run_task_scope.sh` rebuilds the task-class inputs from the ClassEval generations (which are not redistributed) and re-runs the analyses.